

MATHEMATICS

Chapter 1: Relation and Function



RELATIONS AND FUNCTIONS

Top Concepts in Relations

1. Introduction to Relation and no. of relations

- A relation R between two non-empty sets A and B is a subset of their Cartesian product $A \times B$.
- If $A = B$, then the relation R on A is a subset of $A \times A$.
- The total number of relations from a set consisting of m elements to a set consisting of n elements is 2^{mn} .
- If (a, b) belongs to R , then a is related to b and is written as ' $a R b$ '. If (a, b) does not belong to R , then a is not related to b and it is written as ' $a \not R b$ '.

2. Co-domain and Range of a Relation

Let R be a relation from A to B . Then the 'domain of R ' $R' \subset A$ and the 'range of R ' $R'' \subset B$. Co-domain is either set B or any of its superset or subset containing range of R .

3. Types of Relations

A relation R in a set A is called an empty relation if no element of A is related to any element of A , i.e., $R = \phi \subset A \times A$.

A relation R in a set A is called a universal relation if each element of A is related to every element of A , i.e., $R = A \times A$.

4. A relation R on a set A is called:

- Reflexive, if $(a, a) \in R$ for every $a \in A$.
- Symmetric, if $(a_1, a_2) \in R$ implies that $(a_2, a_1) \in R$ for all $a_1, a_2 \in A$.
- Transitive, if $(a_1, a_2) \in R$ and $(a_2, a_3) \in R$ implies that $(a_1, a_3) \in R$ for all $a_1, a_2, a_3 \in A$.

5. Equivalence Relation

- A relation R in a set A is said to be an equivalence relation if R is reflexive, symmetric and transitive.

- An empty relation R on a non-empty set X (i.e., ' $a R b$ ' is never true) is not an equivalence relation, because although it is vacuously symmetric and transitive, but it is not reflexive (except when X is also empty).
6. Given an arbitrary equivalence relation R in a set X , R divides X into mutually disjoint subsets S_i called partitions or subdivisions of X provided:

- All elements of S_i are related to each other for all i .
- No element of S_i is related to any element of S_j if $i \neq j$.

c.
$$\bigcup_{i=1}^n S_i = X \text{ and } S_i \cap S_j = \phi \text{ if } i \neq j.$$

The subsets S_i are called equivalence classes.

7. Union, Intersection and Inverse of Equivalence Relations

- If R and S are two equivalence relations on a set A , $R \cap S$ is also an equivalence relation on A .
- The union of two equivalence relations on a set is not necessarily an equivalence relation on the set.
- The inverse of an equivalence relation is an equivalence relation.

Top Concepts in Functions

1. Introduction to functions

A function from a non-empty set A to another non-empty set B is a correspondence or a rule which associates every element of A to a unique element of B written as $f : A \rightarrow B$ such that $f(x) = y$ for all $x \in A, y \in B$.

All functions are relations, but the converse is not true.

2. Domain, Co-domain and Range of a Function

- If $f : A \rightarrow B$ is a function, then set A is the domain, set B is the co-domain and set $\{f(x) : x \in A\}$ is the range of f .
- The range is a subset of the co-domain.
- A function can also be regarded as a machine which gives a unique output in set B corresponding to each input from set A .

- If A and B are two sets having m and n elements, respectively, then the total number of functions from A to B is n^m .

3. Real Function

- A function $f : A \rightarrow B$ is called a real-valued function if B is a subset of $\mathbb{R}$.
- If A and B both are subsets of $\mathbb{R}$, then 'f' is called a real function.
- While describing real functions using mathematical formula, x (the input) is the independent variable and y (the output) is the dependent variable.
- The graph of a real function 'f' consists of points whose co-ordinates (x, y) satisfy $y = f(x)$, for all $x \in \text{Domain}(f)$.

4. Vertical line test

A curve in a plane represents the graph of a real function if and only if no vertical line intersects it more than once.

5. One-one Function

- A function $f : A \rightarrow B$ is one-to-one if for all $x, y \in A$, $f(x) = f(y) \Rightarrow x = y$ or $x \neq y \Rightarrow f(x) \neq f(y)$.
- A one-one function is known as an injection or injective function. Otherwise, f is called many-one.

6. Onto Function

- A function $f : A \rightarrow B$ is an onto function, if for each $b \in B$, there is at least one $a \in A$ such that $f(a) = b$, i.e., if every element in B is the image of some element in A, then f is an onto or surjective function.
- For an onto function, range = co-domain.
- A function which is both one-one and onto is called a bijective function or a bijection.
- A one-one function defined from a finite set to itself is always onto, but if the set is infinite, then it is not the case.

7. Let A and B be two finite sets and $f : A \rightarrow B$ be a function.

- If f is an injection, then $n(A) \leq n(B)$.
- If f is a surjection, then $n(A) \geq n(B)$.
- If f is a bijection, then $n(A) = n(B)$.

8. If A and B are two non-empty finite sets containing m and n elements, respectively, then

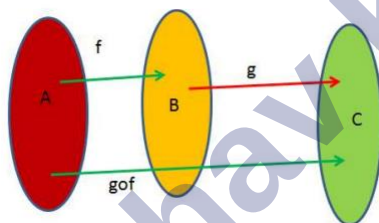
Number of functions from A to B = n^m .

- Number of one-one function from A to B =
$$= \begin{cases} {}^nC_m \times m!, & \text{if } n \geq m \\ 0, & \text{if } n < m \end{cases}$$
- Number of onto functions from A to B =
$$= \begin{cases} \sum_{r=1}^n (-1)^{n-r} {}^nC_r r^m, & \text{if } m \geq n \\ 0, & \text{if } m < n \end{cases}$$
- Number of one-one and onto functions from A to B =
$$= \begin{cases} n!, & \text{if } m=n \\ 0, & \text{if } m \neq n \end{cases}$$

9. If a function $f : A \rightarrow B$ is not an onto function, then $f : A \rightarrow f(A)$ is always an onto function.

10. Composition of Functions

Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be two functions. The composition of f and g , denoted by $g \circ f$, is defined as the function $g \circ f : A \rightarrow C$ and is given by $g \circ f(x) : A \rightarrow C$ defined by $g \circ f(x) = g(f(x)) \forall x \in A$.



- Composition of f and g is written as $g \circ f$ and not $f \circ g$.
- $g \circ f$ is defined if the range of $f \subseteq$ domain of g , and $f \circ g$ is defined if the range of $g \subseteq$ domain of f .
- Composition of functions is not commutative in general i.e., $f \circ g(x) \neq g \circ f(x)$.
- Composition is associative i.e., if $f : X \rightarrow Y$, $g : Y \rightarrow Z$ and $h : Z \rightarrow S$ are functions, then $h \circ (g \circ f) = (h \circ g) \circ f$.
- The composition of two bijections is a bijection.

11. Inverse of a Function

- Let $f : A \rightarrow B$ is a bijection, then $g : B \rightarrow A$ is inverse of f if $f(x) = y \Leftrightarrow g(y) = x$ OR $g \circ f = I_A$ and $f \circ g = I_B$
- If $g \circ f = I_A$ and f is an injection, then g is a surjection.
- If $f \circ g = I_B$ and f is a surjection, then g is an injection.

12. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be two functions. Then

- $g \circ f : A \rightarrow C$ is onto $\Rightarrow g : B \rightarrow C$ is onto.
- $g \circ f : A \rightarrow C$ is one-one $\Rightarrow f : A \rightarrow B$ is one-one.

- $g \circ f: A \rightarrow C$ is onto and $g: B \rightarrow C$ is one-one $\Rightarrow f: A \rightarrow B$ is onto.
- $g \circ f: A \rightarrow C$ is one-one and $f: A \rightarrow B$ is onto $\Rightarrow g: B \rightarrow C$ is one-one.

13. Invertible Function

- A function $f: X \rightarrow Y$ is defined to be invertible if there exists a function $g: Y \rightarrow X$ such that $g \circ f = I_X$ and $f \circ g = I_Y$.
- The function g is called the inverse of f and is denoted by f^{-1} . If f is invertible, then f must be one-one and onto, and conversely, if f is one-one and onto, then f must be invertible.
- If $f: A \rightarrow B$ and $g: B \rightarrow C$ are one-one and onto, then $g \circ f: A \rightarrow C$ is also one-one and onto. But if $g \circ f$ is one-one, then only f is one-one and g may or may not be one-one. If $g \circ f$ is onto, then g is onto and f may or may not be onto.
- Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be two invertible functions. Then $g \circ f$ is also invertible with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
- If $f: R \rightarrow R$ is invertible, $f(x) = y$, then $f^{-1}(y) = x$ and $(f^{-1})^{-1}$ is the function f itself.

Binary Operations

1. A binary operation $*$ on a set A is a function from $A \times A$ to A .
2. If $*$ is a binary operation on a set S , then S is closed with respect to $*$.

3. Binary operations on R

- Addition, subtraction and multiplication are binary operations on R , which is the set of real numbers.
- Division is not binary on R ; however, division is a binary operation on $R - \{0\}$ which is the set of non-zero real numbers.

4. Laws of Binary Operations

- A binary operation $*$ on the set X is called commutative, if $a * b = b * a$, for every $a, b \in X$.
- A binary operation $*$ on the set X is called associative, if $a(b * c) = (a * b) * c$, for every $a, b, c \in X$.
- An element $e \in A$ is called an identity of A with respect to $*$ if for each $a \in A$, $a * e = a = e * a$.
- The identity element of $(A, *)$ if it exists, is unique.

5. Existence of Inverse

Given a binary operation $*$ from $A \times A \rightarrow A$ with the identity element e in A , an element $a \in A$ is said to be invertible with respect to the operation $*$, if there exists an element b in

A such that $a * b = e = b * a$ and b is called the inverse of a and is denoted by a^{-1} .

6. If the operation table is symmetric about the diagonal line, then the operation is commutative.

*	a	b	c	d
a	a	b	c	d
b	b	c	d	a
c	c	d	a	b
d	d	a	b	c

The operation $*$ is commutative.

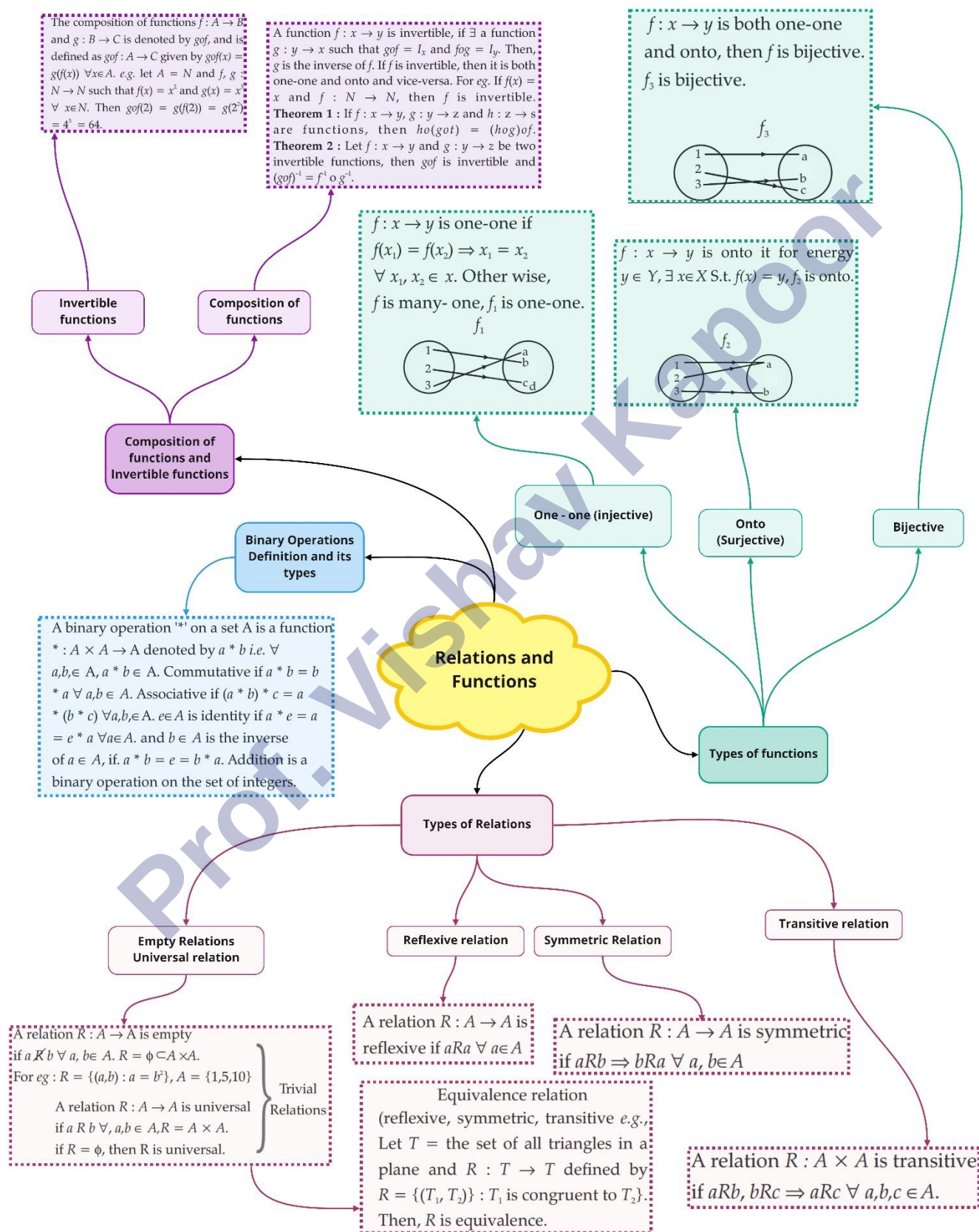
7. Binary Operation on Natural Numbers

Addition '+' and multiplication '-' on N , the set of natural numbers, are binary operations. However, subtraction '-' and division are not, because $(4, 5) = 4 - 5 = -1 \notin N$ and $4/5 = .8 \notin N$.

8. Number of Binary Operations

- Let S be a finite set consisting of n elements. Then $S \times S$ has n^2 elements.
- The total number of functions from a finite set A to a finite set B is $[n(B)]^{n(A)}$. Therefore, total number of binary operations on S is n^{n^2} .
- The total number of commutative binary operations on a set consisting of n elements is $n \frac{n(n-1)}{2}$.

Class : 12th Maths
Chapter- 1 : Relations and Functions



Important Questions

Multiple Choice questions-

1. Let R be the relation in the set $\{1, 2, 3, 4\}$, given by:

$$R = \{(1, 2), (2, 2), (1, 1), (4, 4), (1, 3), (3, 3), (3, 2)\}.$$

Then:

- (a) R is reflexive and symmetric but not transitive
- (b) R is reflexive and transitive but not symmetric
- (c) R is symmetric and transitive but not reflexive
- (d) R is an equivalence relation.

2. Let R be the relation in the set N given by: $R = \{(a, b) : a = b - 2, b > 6\}$. Then:

- (a) $(2, 4) \in R$
- (b) $(3, 8) \in R$
- (c) $(6, 8) \in R$
- (d) $(8, 7) \in R$.

3. Let $A = \{1, 2, 3\}$. Then number of relations containing $\{1, 2\}$ and $\{1, 3\}$, which are reflexive and symmetric but not transitive is:

- (a) 1
- (b) 2
- (c) 3
- (d) 4.

4. Let $A = \{1, 2, 3\}$. Then the number of equivalence relations containing $(1, 2)$ is

- (a) 1
- (b) 2
- (c) 3
- (d) 4.

5. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^4$. Then

- (a) f is one-one onto
- (b) f is many-one onto
- (c) f is one-one but not onto
- (d) f is neither one-one nor onto.

6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = 3x$. Then

- (a) f is one-one onto
- (b) f is many-one onto
- (c) f is one-one but not onto
- (d) f is neither one-one nor onto.

7. If $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = (3 - x^3)^{1/3}$, then $f \circ f(x)$ is

- (a) $x^{1/3}$
- (b) x^3
- (c) x
- (d) $3 - x^3$.

8. Let $f: \mathbb{R} - \{-\frac{4}{3}\} \rightarrow \mathbb{R}$ be a function defined as: $f(x) = \frac{4x}{3x+4}$, $x \neq -\frac{4}{3}$. The inverse of f is map $g: \text{Range } f \rightarrow \mathbb{R} - \{-\frac{4}{3}\}$ given by

- (a) $g(y) = \frac{3y}{3-4y}$
- (b) $g(y) = \frac{4y}{4-3y}$
- (c) $g(y) = \frac{4y}{3-4y}$
- (d) $g(y) = \frac{3y}{4-3y}$

9. Let R be a relation on the set $\mathbb{N}$ of natural numbers defined by nRm if n divides m . Then R is

- (a) Reflexive and symmetric

- (b) Transitive and symmetric
- (c) Equivalence
- (d) Reflexive, transitive but not symmetric.

10. Set A has 3 elements, and the set B has 4 elements. Then the number of injective mappings that can be defined from A to B is:

- (a) 144
- (b) 12
- (c) 24
- (d) 64

Very Short Questions:

1. If $R = \{(x, y) : x + 2y = 8\}$ is a relation in N , write the range of R .
2. Show that a one-one function:
 $f : \{1, 2, 3\} \rightarrow \{1, 2, 3\}$ must be onto. (N.C.E.R.T.)
3. What is the range of the function $f(x) = \frac{|x-1|}{x-1}$? (C.B.S.E. 2010)
4. Show that the function $f : N \rightarrow N$ given by $f(x) = 2x$ is one-one but not onto. (N.C.E.R.T.)
5. If $f : R \rightarrow R$ is defined by $f(x) = 3x + 2$ find $f(f(x))$. C.B.S.E. 2011 (F)
6. If $f(x) = \frac{x}{x-1}$, $x \neq 1$ then find $f \circ f$. (N.C.E.R.T.)
7. If $f : R \rightarrow R$ is defined by $f(x) = (3 - x^3)^{1/3}$, find $f \circ f(x)$
8. Are f and g both necessarily onto, if $g \circ f$ is onto? (N.C.E.R.T.)

Short Questions:

1. Let A be the set of all students of a Boys' school. Show that the relation R in A given by:

$R = \{(a, b) : a \text{ is sister of } b\}$ is an empty relation and the relation R' given by :

$R' = \{(a, b) : \text{the difference between heights of } a \text{ and } b \text{ is less than 3 metres}\}$ is an universal relation. (N.C.E.R.T.)

2. Let $f : X \rightarrow Y$ be a function. Define a relation R in X given by :

$$R = \{(a,b):f(a) = f(b)\}.$$

Examine, if R is an equivalence relation. (N.C.E.R.T.)

3. Let R be the relation in the set Z of integers given by:

$$R = \{(a, b): 2 \text{ divides } a - b\}.$$

Show that the relation R is transitive. Write the equivalence class $[0]$. (C.B.S.E. Sample Paper 2019-20)

4. Show that the function:

$$f : \mathbb{N} \rightarrow \mathbb{N}$$

given by $f(1) = f(2) = 1$ and $f(x) = x - 1$, for every $x > 2$ is onto but not one-one. (N.C.E.R.T.)

5. Find gof and fog , if:

$f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are given by $f(x) = \cos x$ and $g(x) = 3x^2$. Show that $\text{gof} \neq \text{fog}$. (N. C.E.R. T.)

6. If $f(x) = \frac{4x+3}{6x-4}$, $x \neq \frac{2}{3}$ find $\text{fof}(x)$

7. Let $A = \mathbb{N} \times \mathbb{N}$ be the set of all ordered pairs of natural numbers and R be the relation on the set A defined by $(a, b) R (c, d)$ iff $ad = bc$. Show that R is an equivalence relation.

8. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the Signum function defined as:

$$f(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

and $g : \mathbb{R} \rightarrow \mathbb{R}$ be the Greatest Integer Function given by $g(x) = [x]$, where $[x]$ is greatest integer less than or equal to x . Then does fog and gof coincide in $(0,1]$?

Long Questions:

1. Show that the relation R on $\mathbb{R}$ defined as $R = \{(a, b): a \leq b\}$, is reflexive and transitive but not symmetric.

2. Prove that function $f : \mathbb{N} \rightarrow \mathbb{N}$, defined by $f(x) = x^2 + x + 1$ is one-one but not onto. Find inverse of $f : \mathbb{N} \rightarrow S$, where S is range of f .
3. Let $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$.

Show that $R = \{(a, b) : a, b \in A; |a - b| \text{ is divisible by } 4\}$ is an equivalence relation. Find the set of all elements related to 1. Also write the equivalence class $[2]$. (C.B.S.E 2018)

4. Prove that the function $f: [0, \infty) \rightarrow \mathbb{R}$ given by $f(x) = 9x^2 + 6x - 5$ is not invertible. Modify the co-domain of the function f to make it invertible, and hence find f^{-1} . (C.B.S.E. Sample Paper 2018-19)

Assertion and Reason Questions-

1. Two statements are given-one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to these questions from the codes(a), (b), (c) and (d) as given below.

- a) Both A and R are true and R is the correct explanation of A.
- b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false.
- d) A is false and R is also false.

Assertion(A): Let L be the set of all lines in a plane and R be the relation in L defined as $R = \{(L_1, L_2) : L_1 \text{ is perpendicular to } L_2\}$. R is not equivalence relation.

Reason (R): R is symmetric but neither reflexive nor transitive

2. Two statements are given-one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to these questions from the codes(a), (b), (c) and (d) as given below.

- a) Both A and R are true and R is the correct explanation of A.
- b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false.
- d) A is false and R is also false.

Assertion (A): $R = \{(T_1, T_2) : T_1 \text{ is congruent to } T_2\}$. Then R is an equivalence relation.

Reason(R): Any relation R is an equivalence relation, if it is reflexive, symmetric and transitive.

Case Study Questions-

1. Consider the mapping $f: A \rightarrow B$ is defined by $f(x) = x - 1$ such that f is a bijection.

Based on the above information, answer the following questions.

(i) Domain of f is:

- a) $\mathbb{R} - \{2\}$
- b) $\mathbb{R}$
- c) $\mathbb{R} - \{1, 2\}$
- d) $\mathbb{R} - \{0\}$

(ii) Range of f is:

- a) $\mathbb{R}$
- b) $\mathbb{R} - \{2\}$
- c) $\mathbb{R} - \{0\}$
- d) $\mathbb{R} - \{1, 2\}$

(iii) If $g: \mathbb{R} - \{2\} \rightarrow \mathbb{R} - \{1\}$ is defined by $g(x) = 2f(x) - 1$, then $g(x)$ in terms of x is:

- a. $\frac{x+2}{x}$
- b. $\frac{x+1}{x-2}$
- c. $\frac{x-2}{x}$
- d. $\frac{x}{x-2}$

(iv) The function g defined above, is:

- a) One-one
- b) Many-one
- c) into
- d) None of these

(v) A function $f(x)$ is said to be one-one if.

- a. $f(x_1) = f(x_2) \Rightarrow -x_1 = x_2$
- b. $f(-x_1) = f(-x_2) \Rightarrow -x_1 = x_2$
- c. $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$
- d. None of these

2. A relation R on a set A is said to be an equivalence relation on A iff it is:

- I. Reflexive i.e., $(a, a) \in R \forall a \in A$.
- II. Symmetric i.e., $(a, b) \in R \Rightarrow (b, a) \in R \forall a, b \in A$.
- III. Transitive i.e., $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R \forall a, b, c \in A$.

Based on the above information, answer the following questions.

(i) If the relation $R = \{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$ defined on the set $A = \{1, 2, 3\}$, then R is:

- a) Reflexive
- b) Symmetric
- c) Transitive
- d) Equivalence

(ii) If the relation $R = \{(1, 2), (2, 1), (1, 3), (3, 1)\}$ defined on the set $A = \{1, 2, 3\}$, then R is:

- a) Reflexive
- b) Symmetric
- c) Transitive
- d) Equivalence

(iii) If the relation R on the set N of all natural numbers defined as $R = \{(x, y): y = x + 5 \text{ and } x < 4\}$, then R is:

- a) Reflexive
- b) Symmetric
- c) Transitive
- d) Equivalence

(iv) If the relation R on the set $A = \{1, 2, 3, \dots, 13, 14\}$ defined as $R = \{(x, y): 3x - y = 0\}$, then R is:

- a) Reflexive
- b) Symmetric
- c) Transitive
- d) Equivalence

(v) If the relation R on the set $A = \{1, 2, 3\}$ defined as $R = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$, then R is:

- a) Reflexive only

- b) Symmetric only
- c) Transitive only
- d) Equivalence

Answer Key-

Multiple Choice questions-

(b) R is reflexive and transitive but not symmetric

(c) $(6, 8) \in R$

(a) 1

(b) 2

(d) f is neither one-one nor onto.

(a) f is one-one onto

(c) x

(b) $g(y) = \frac{4y}{4-3y}$

(b) Transitive and symmetric

(c) 24

Very Short Answer:

1. Solution: Range of $R = \{1, 2, 3\}$.

[$\because$ When $x = 2$, then $y = 3$, when $x = 4$, then $y = 2$, when $x = 6$, then $y = 1$]

2. Solution: Since 'f' is one-one,

$\therefore$ under 'f', all the three elements of $\{1, 2, 3\}$ should correspond to three different elements of the co-domain $\{1, 2, 3\}$.

Hence, 'f' is onto.

3. Solution: When $x > 1$,

$$\text{then } f(x) = \frac{x-1}{x-1} = 1.$$

When $x < 1$,

$$\text{than } f(x) = \frac{-(x-1)}{x-1} = -1$$

$$\text{Hence, } R_f = \{-1, 1\}.$$

4. Solution:

$$\text{Let } x_1, x_2 \in \mathbb{N}.$$

$$\text{Now, } f(x_1) = f(x_2)$$

$$\Rightarrow 2x_1 = 2x_2$$

$$\Rightarrow x_1 = x_2$$

$$\Rightarrow f \text{ is one-one.}$$

$$\text{Now, } f \text{ is not onto.}$$

$$\because \text{For } 1 \in \mathbb{N}, \text{ there does not exist any } x \in \mathbb{N} \text{ such that } f(x) = 2x = 1.$$

$$\text{Hence, } f \text{ is one-one but not onto.}$$

5. Solution:

$$f(f(x)) = 3f(x) + 2$$

$$= 3(3x + 2) + 2 = 9x + 8.$$

6. Solution:

$$\begin{aligned} f \circ f(x) &= f(f(x)) = \frac{f(x)}{f(x)-1} \\ &= \frac{\frac{x}{x-1}}{\frac{x}{x-1}-1} = \frac{x}{x-x+1} \\ &= \frac{x}{1} = x. \end{aligned}$$

7. Solution:

$$f \circ f(x) = f(f(x)) = (3 - (f(x))^3)^{1/3}$$

$$= (3 - ((3 - x^3)^{1/3})^3)^{1/3}$$

$$= (3 - (3 - x^3))^{1/3} = (x^3)^{1/3} = x.$$

8. Solution:

Consider $f: \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4\}$

and $g: \{1, 2, 3, 4\} \rightarrow \{1, 2, 3\}$ defined by:

$$f(1) = 1, f(2) = 2, f(3) = f(4) = 3$$

$$g(1) = 1, g(2) = 2, g(3) = g(4) = 3.$$

$\therefore \text{gof} = g(f(x)) \{1, 2, 3\}$, which is onto

But f is not onto.

$[\because 4$ is not the image of any element]

Short Answer:

1. Solution:

(i) Here $R = \{(a, b): a \text{ is sister of } b\}$.

Since the school is a Boys' school,

$\therefore$ no student of the school can be the sister of any student of the school.

Thus $R = \Phi$ Hence, R is an empty relation.

(ii) Here $R' = \{(a, b): \text{the difference between heights of } a \text{ and } b \text{ is less than } 3 \text{ metres}\}$.

Since the difference between heights of any two students of the school is to be less than 3 metres,

$\therefore R' = A \times A$. Hence, R' is a universal relation.

2. Solution:

For each $a \in X$, $(a, a) \in R$.

Thus R is reflexive. $[\because f(a) = f(a)]$

Now $(a, b) \in R$

$$\Rightarrow f(a) = f(b)$$

$$\Rightarrow f(b) = f(a)$$

$$\Rightarrow (b, a) \in R.$$

Thus R is symmetric.

And $(a, b) \in R$

and $(b, c) \in R$

$$\Rightarrow f(a) = f(b)$$

$$\text{and } f(b) = f(c)$$

$$\Rightarrow f(a) = f(c)$$

$$\Rightarrow (a, c) \in R.$$

Thus R is transitive.

Hence, R is an equivalence relation.

3. Solution:

Let 2 divide $(a - b)$ and 2 divide $(b - c)$, where $a, b, c \in \mathbb{Z}$

$$\Rightarrow 2 \text{ divides } [(a - b) + (b - c)]$$

$$\Rightarrow 2 \text{ divides } (a - c).$$

Hence, R is transitive.

$$\text{And } [0] = \{0, \pm 2, \pm 4, \pm 6, \dots\}.$$

4. Solution:

$$\text{Since } f(1) = f(2) = 1,$$

$$\therefore f(1) = f(2), \text{ where } 1 \neq 2.$$

$$\therefore 'f' \text{ is not one-one.}$$

$$\text{Let } y \in \mathbb{N}, y \neq 1,$$

$$\text{we can choose } x \text{ as } y + 1 \text{ such that } f(x) = x - 1$$

$$= y + 1 - 1 = y.$$

$$\text{Also } 1 \in \mathbb{N}, f(1) = 1.$$

Thus ' f ' is onto.

Hence, ' f ' is onto but not one-one.

5. Solution:

We have:

$$f(x) = \cos x \text{ and } g(x) = 3x^2.$$

$$\therefore \text{gof}(x) = g(f(x)) = g(\cos x)$$

$$= 3(\cos x)^2 = 3 \cos^2 x$$

$$\text{and } \text{fog}(x) = f(g(x)) = f(3x^2) = \cos 3x^2.$$

Hence, $\text{gof} \neq \text{fog}$.

6. Solution:

$$\text{We have: } \frac{4x+3}{6x-4} \dots (1)$$

$$\therefore \text{fof}(x) = f(f(x))$$

$$= \frac{4f(x)+3}{6f(x)-4}$$

$$= \frac{4\left(\frac{4x+3}{6x-4}\right)+3}{6\left(\frac{4x+3}{6x-4}\right)-4} \quad [\text{Using (1)}]$$

$$= \frac{16x+12+18x-12}{24x+18-24x+16}$$

$$= \frac{34x}{34} = x.$$

7. Solution:

Given: $(a, b) R (c, d)$ if and only if $ad = bc$.

(I) $(a, b) R (a, b)$ iff $ab = ba$, which is true.

$$[\because ab = ba \forall a, b \in \mathbb{N}]$$

Thus, R is reflexive.

(II) $(a, b) R (c, d) \Rightarrow ad = bc$

$$(c, d) R (a, b) \Rightarrow cb = da.$$

But $cb = be$ and $da = ad$ in N .

$$\therefore (a, b) R (c, d) \Rightarrow (c, d) R (a, b).$$

Thus, R is symmetric.

$$(III) (a, b) R (c, d)$$

$$\Rightarrow ad = bc \dots (1)$$

$$(c, d) R (e, f)$$

$$\Rightarrow cf = de \dots (2)$$

Multiplying (1) and (2), $(ad) \cdot (cf) = (be) \cdot (de)$

$$\Rightarrow af = be$$

$$\Rightarrow (a, b) R (e, f).$$

Thus, R is transitive.

Thus, R is reflexive, symmetric and transitive.

Hence, R is an equivalence relation.

8. Solution:

For $x \in [0, 1]$.

$$(f \circ g)(x) = f(g(x)) = f([x])$$

$$= \begin{cases} f(0); & \text{if } 0 < x < 1 \\ f(1); & \text{if } x = 1 \end{cases}$$

$$\Rightarrow f(g(x)) = \begin{cases} 0; & \text{if } 0 < x < 1 \\ 1; & \text{if } x = 1 \end{cases} \dots (1)$$

$$\text{And } (g \circ f)(x) = g(f(x)) = g(1)$$

$$[\because f(x) = 1 \forall x > 0]$$

$$= [1] = 1$$

$$\Rightarrow (g \circ f)(x) = 1 \forall x \in [0, 1] \dots (2)$$

From (1) and (2), $(f \circ g)$ and $(g \circ f)$ do not coincide in $(0, 1]$.

Long Answer:

1. Solution:

We have: $R = \{(a, b) \mid a \leq b\}$.

Since, $a \leq a \forall a \in R$,

$\therefore (a, a) \in R$,

Thus, R reflexive.

Now, $(a, b) \in R$ and $(b, c) \in R$

$\Rightarrow a \leq b$ and $b \leq c$

$\Rightarrow a \leq c$

$\Rightarrow (a, c) \in R$.

Thus, R is transitive.

But R is not symmetric

$[\because (3, 5) \in R$ but $(5, 3) \notin R$ as $3 \leq 5$ but $5 > 3]$

Solution:

Let $x_1, x_2 \in \mathbb{N}$.

Now, $f(x_1) = f(x_2)$

$$\Rightarrow x_1^2 + x_1 + 1 = x_2^2 + x_2 + 1$$

$$\Rightarrow x_1^2 + x_1 = x_2^2 + x_2$$

$$\Rightarrow (x_1^2 - x_2^2) + (x_1 - x_2) = 0$$

$$\Rightarrow (x_1 - x_2) + (x_1 + x_2 + 1) = 0$$

$$\Rightarrow x_1 - x_2 = 0 \quad [\because x_1 + x_2 + 1 \neq 0]$$

$$\Rightarrow x_1 = x_2.$$

Thus, f is one-one.

Let $y \in \mathbb{N}$, then for any x ,

$$f(x) = y \text{ if } y = x^2 + x + 1$$

$$\Rightarrow y = \left(x^2 + x + \frac{1}{4}\right) + \frac{3}{4}$$

$$\Rightarrow y = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$$

$$\Rightarrow x + \frac{1}{2} = \pm \sqrt{y - \frac{3}{4}}$$

$$\Rightarrow x = \pm \frac{\sqrt{4y-3}}{2} - \frac{1}{2}$$

$$\Rightarrow x = \frac{\pm\sqrt{4y-3}-1}{2}$$

$$\Rightarrow x = \frac{\sqrt{4y-3}-1}{2}$$

$$\left[\frac{-\sqrt{4y-3}-1}{2} \notin N \text{ for any value of } y \right]$$

Now, for $y = \frac{3}{4}$, $x = -\frac{1}{2} \notin N$.

Thus, f is not onto.

$\Rightarrow f(x)$ is not invertible.

Since, $x > 0$, therefore, $\frac{\sqrt{4y-3}-1}{2} > 0$

$$\Rightarrow \sqrt{4y-3} > 1$$

$$\Rightarrow 4y - 3 > 1$$

$$\Rightarrow 4y > 4$$

$$\Rightarrow y > 1.$$

Redefining, $f : (0, \infty) \rightarrow (1, \infty)$ makes

$f(x) = x^2 + x + 1$ on onto function.

Thus, $f(x)$ is bijection, hence f is invertible and $f^{-1} : (1, \infty) \rightarrow (0, \infty)$

$$f^{-1}(y) = \frac{\sqrt{4y-3}-1}{2}$$

2. Solution:

We have:

$R = \{(a, b) : a, b \in A; |a - b| \text{ is divisible by } 4\}.$

(1) Reflexive: For any $a \in A$,

$\therefore (a, a) \in R.$

$|a - a| = 0$, which is divisible by 4.

Thus, R is reflexive.

Symmetric:

Let $(a, b) \in R$

$\Rightarrow |a - b|$ is divisible by 4

$\Rightarrow |b - a|$ is divisible by 4

Thus, R is symmetric.

Transitive: Let $(a, b) \in R$ and $(b, c) \in R$

$\Rightarrow |a - b|$ is divisible by 4 and $|b - c|$ is divisible by 4

$\Rightarrow |a - b| = 4\lambda$

$\Rightarrow a - b = \pm 4\lambda \dots\dots\dots(1)$

and $|b - c| = 4\mu$

$\Rightarrow b - c = \pm 4\mu \dots\dots\dots(2)$

Adding (1) and (2),

$(a - b) + (b - c) = \pm 4(\lambda + \mu)$

$\Rightarrow a - c = \pm 4(\lambda + \mu)$

$\Rightarrow (a, c) \in R.$

Thus, R is transitive.

Now, R is reflexive, symmetric and transitive.

Hence, R is an equivalence relation.

(ii) Let 'x' be an element of A such that $(x, 1) \in R$

$\Rightarrow |x - 1|$ is divisible by 4

$$\Rightarrow x - 1 = 0, 4, 8, 12, \dots$$

$$\Rightarrow x = 1, 5, 9, 13, \dots$$

Hence, the set of all elements of A which are related to 1 is $\{1, 5, 9\}$.

(iii) Let $(x, 2) \in R$.

Thus $|x - 2| = 4k$, where $k \leq 3$.

$$\therefore x = 2, 6, 10.$$

Hence, equivalence class $[2] = \{2, 6, 10\}$.

3. Solution:

Let $y \in \mathbb{R}$.

For any x , $f(x) = y$ if $y = 9x^2 + 6x - 5$

$$\Rightarrow y = (9x^2 + 6x + 1) - 6$$

$$= (3x + 1)^2 - 6$$

$$\Rightarrow 3x + 1 = \pm\sqrt{y+6}$$

$$\Rightarrow x = \frac{\pm\sqrt{y+6}-1}{3}$$

$$\Rightarrow x = \frac{\sqrt{y+6}-1}{3}$$

$$\left[\because \frac{-\sqrt{y+6}-1}{3} \notin [0, \infty) \text{ for any value of } y \right]$$

For $y = -6 \in \mathbb{R}$, $x = -\frac{1}{3} \notin [0, \infty)$.

Thus, $f(x)$ is not onto.

Hence, $f(x)$ is not invertible.

$$\text{Since, } x \geq 0, \therefore \frac{\sqrt{y+6}-1}{3} \geq 0$$

$$\Rightarrow \sqrt{y+6}-1 \geq 0$$

$$\Rightarrow \sqrt{y+6} \geq 1$$

$$\Rightarrow y+6 \geq 1$$

$$\Rightarrow y \geq -5.$$

We redefine,

$$f: [0, \infty) \rightarrow [-5, \infty),$$

which makes $f(x) = 9x^2 + 6x - 5$ an onto function.

Now, $x_1, x_2 \in [0, \infty)$ such that $f(x_1) = f(x_2)$

$$\Rightarrow (3x_1 + 1)^2 = (3x_2 + 1)^2$$

$$\Rightarrow [(3x_1 + 1) + (3x_2 + 1)][(3x_1 + 1) - (3x_2 + 1)]$$

$$\Rightarrow [3(x_1 + x_2) + 2][3(x_1 - x_2)] = 0$$

$$\Rightarrow x_1 = x_2$$

$$[\because 3(x_1 + x_2) + 2 > 0]$$

Thus, $f(x)$ is one-one.

$\therefore f(x)$ is bijective, hence f is invertible

$$\text{and } f^{-1}: [-5, \infty) \rightarrow [0, \infty)$$

$$f^{-1}(y) = \frac{\sqrt{y+6}-1}{3}$$

Assertion and Reason Answers-

1. (a) Both A and R are true and R is the correct explanation of A.
2. (a) Both A and R are true and R is the correct explanation of A.

Case Study Answers-

1. Answer :

$$(i) (a) \mathbb{R} - \{2\}$$

Solution:

For $f(x)$ to be defined $x - 2; \neq 0$ i.e., $x; \neq 2$.

$$\therefore \text{Domain of } f = \mathbb{R} - \{2\}$$

(ii) (b) $R - \{2\}$ **Solution:**Let $y = f(x)$, then $y = \frac{x-1}{x-2}$

$$\Rightarrow xy - 2y = x - 1 \Rightarrow xy - x = 2y - 1$$

$$\Rightarrow x = \frac{2y-1}{y-1}$$

Since, $x \in R - \{2\}$, therefore $y \neq 1$ Hence, range of $f = R - \{1\}$ (iii) (d) $\frac{x}{x-2}$ **Solution:**We have, $g(x) = 2f(x) - 1$

$$= 2\left(\frac{x-1}{x-2}\right) - 1 = \frac{2x-2-x+2}{x-2} = \frac{x}{x-2}$$

(iv) (a) One-one

Solution:We have, $g(x) = \frac{x}{x-2}$

$$\text{Let } g(x_1) = g(x_2) \Rightarrow \frac{x_1}{x_1-2} = \frac{x_2}{x_2-2}$$

$$\Rightarrow x_1x_2 - 2x_1 = x_1x_2 - 2x_2 \Rightarrow 2x_1 = 2x_2 \Rightarrow x_1 = x_2$$

$$\text{Thus, } g(x_1) = g(x_2) \Rightarrow x_1 = x_2$$

Hence, $g(x)$ is one-one.

$$(v)(c) f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$

2. Answer :

(i) (a) Reflexive

Solution:

Clearly, $(1, 1), (2, 2), (3, 3), \in R$. So, R is reflexive on A .

Since, $(1, 2) \in R$ but $(2, 1) \notin R$. So, R is not symmetric on A .

Since, $(2, 3), \in R$ and $(3, 1) \in R$ but $(2, 1) \notin R$. So, R is not transitive on A .

(ii) (b) Symmetric

Solution:

Since, $(1, 1), (2, 2)$ and $(3, 3)$ are not in R . So, R is not reflexive on A .

Now, $(1, 2) \in R \Rightarrow (2, 1) \in R$ and $(1, 3) \in R \Rightarrow (3, 1) \in R$. So, R is symmetric,

Clearly, $(1, 2) \in R$ and $(2, 1) \in R$ but $(1, 1) \notin R$. So, R is not transitive on A .

(iii) (c) Transitive

Solution:

We have, $R = \{(x, y): y = x + 5 \text{ and } x < 4\}$, where $x, y \in \mathbb{N}$.

$$\therefore R = \{(1, 6), (2, 7), (3, 8)\}$$

Clearly, $(1, 1), (2, 2)$ etc. are not in R . So, R is not reflexive.

Since, $(1, 6) \in R$ but $(6, 1) \notin R$. So, R is not symmetric.

Since, $(1, 6) \in R$ and there is no order pair in R which has 6 as the first element.

Same is the case for $(2, 7)$ and $(3, 8)$. So, R is transitive.

(iv) (d) Equivalence

Solution:

We have, $R = \{(x, y): 3x - y = 0\}$, where $x, y \in A = \{1, 2, \dots, 14\}$.

$$\therefore R = \{(1, 3), (2, 6), (3, 9), (4, 12)\}$$

Clearly, $(1, 1) \notin R$. So, R is not reflexive on A .

Since, $(1, 3) \in R$ but $(3, 1) \notin R$. So, R is not symmetric on A .

Since, $(1, 3) \in R$ and $(3, 9) \in R$ but $(1, 9) \notin R$. So, R is not transitive on A .

(v)(d) Equivalence

Solution:

Clearly, $(1, 1), (2, 2), (3, 3) \in R$. So, R is reflexive on A .

We find that the ordered pairs obtained by interchanging the components of ordered pairs in R are also in R . So, R is symmetric on A . For $1, 2, 3 \in A$ such that $(1, 2)$ and $(2, 3)$ are in R implies that $(1, 3)$ is also, in R . So, R is transitive on A . Thus, R is an equivalence relation.

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